

AI Interview Analyzer Pro

Academic Deep Learning Project Report

Project Type	Multimodal AI Interview-Practice Analyzer
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Version	AI_Interview_Analyzer_Pro_Academic
Deployment Target	Streamlit Community Cloud and Local Python Environment
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This report documents the design, implementation, deployment, testing, ethical constraints, and academic contribution of an AI-based interview analysis system. The system accepts a recorded interview video and produces a practice-oriented dashboard with speech transcription, voice-delivery indicators, visual-attention estimation, language analysis, scoring, and downloadable reports.

Important Responsible-Use Note

The system is designed for educational demonstration and self-improvement only. It must not be used as a final hiring, rejection, grading, medical, psychological, honesty, personality, intelligence, or competence decision system. Human review is required for any real evaluation.

Abstract

AI Interview Analyzer Pro is an academic multimodal deep learning and machine learning project that analyzes an interview recording from multiple perspectives: facial emotion patterns, camera-facing attention, speech-to-text transcription, acoustic voice characteristics, sentiment, filler words, keyword coverage, and final practice scoring. The application is implemented as a Streamlit dashboard so that users can upload short interview videos and receive interactive results, charts, transcript downloads, score summaries, and a PDF report.

The full academic version was designed to use DeepFace for emotion recognition, OpenCV and MediaPipe-style logic for visual attention, Faster-Whisper for automatic speech recognition, Librosa for audio feature extraction, TextBlob for sentiment, and ReportLab for PDF reporting. During public cloud deployment, a lightweight configuration was also prepared because TensorFlow, DeepFace, and desktop OpenCV dependencies can exceed the free memory and Linux dependency limits of Streamlit Community Cloud. Therefore, the deployed cloud version is optimized for reliability, while the full emotion-recognition version can be demonstrated locally.

1. Introduction

Interview preparation is a common requirement for students, graduates, and professionals applying for jobs, internships, scholarships, and academic opportunities. Traditional mock interviews usually rely on human feedback, which can be helpful but time-consuming and inconsistent. With the growth of artificial intelligence, it is possible to build supportive practice systems that provide structured feedback on voice delivery, speech clarity, body presence, and answer content.

This project develops a practical AI-powered interview analyzer that allows a candidate to upload a short interview video and receive immediate feedback. The goal is not to replace a human interviewer. Instead, the system acts as an interview-practice assistant that highlights areas such as speaking pace, filler words, transcript quality, camera-facing attention, and expected keyword coverage.

2. Problem Statement

Many learners and early-career professionals struggle to evaluate their interview performance objectively. They may not know whether their answer contains relevant keywords, whether they speak too fast or too slowly, whether their audio is clear, or whether they maintain a reasonable camera-facing posture. Human feedback is valuable, but it is not always available on demand. The project addresses this gap by developing a multimodal AI dashboard that transforms an interview recording into structured, measurable, and downloadable practice feedback.

3. Project Objectives

- Develop a Streamlit-based web application where a user can upload an interview video.
- Extract audio from the video and convert it into a standard WAV format for speech and voice processing.
- Generate a transcript using an automatic speech-recognition model.
- Analyze the transcript for sentiment, filler words, vocabulary diversity, and expected keyword coverage.
- Extract voice-delivery features such as speaking rate, loudness proxy, pitch proxy, and silence ratio.
- Estimate camera-facing attention using computer vision and face-detection logic.
- Provide a weighted overall practice score with individual sub-scores for different interview dimensions.
- Generate downloadable CSV and PDF reports suitable for academic submission and portfolio demonstration.
- Document system limitations and responsible-use constraints to avoid misuse as an automated hiring decision system.

4. Scope of the Project

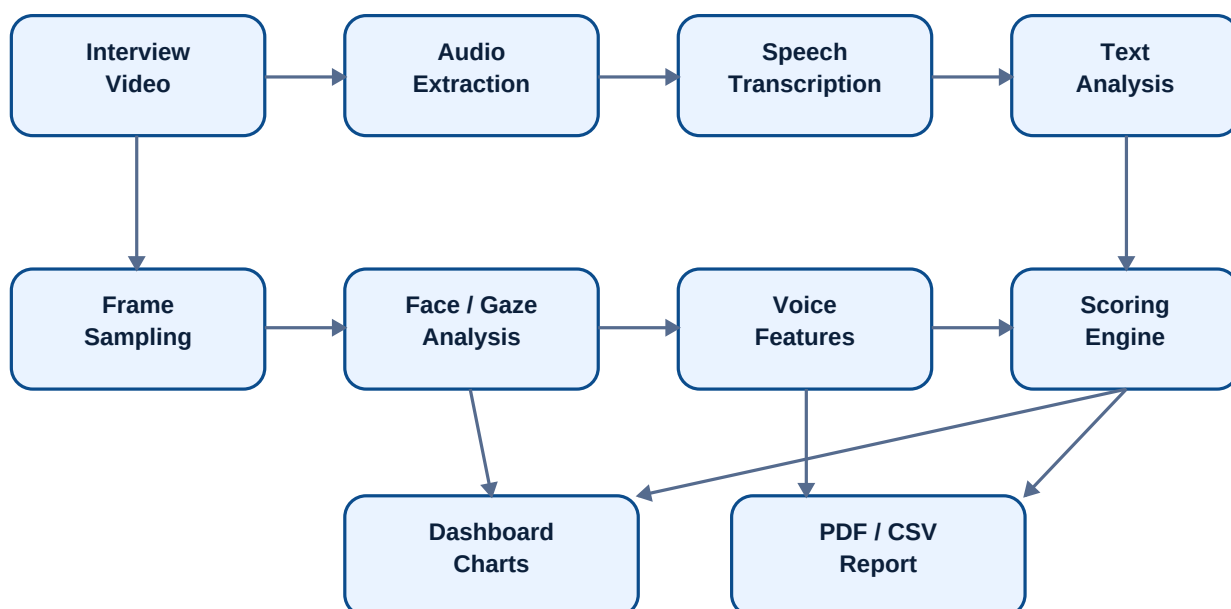
The system is designed for short practice interviews, typically between 20 seconds and 2 minutes for cloud use and longer videos for local demonstrations. It accepts common video formats such as MP4, MOV, AVI, and MKV. The application focuses on interview practice and academic demonstration, not commercial hiring automation. The

primary users are students, trainees, and instructors who want a structured view of interview performance.

Included in Scope	Not Included in Scope
Video upload and playback	Automated hiring or rejection decisions
Audio extraction and speech transcription	Psychological diagnosis or personality judgment
Voice-delivery and transcript analysis	Guaranteed emotion or intention detection
Camera-facing attention proxy	Clinical eye tracking
PDF and CSV report generation	Processing confidential interviews without consent
Local full version and cloud-friendly version	Unlimited long-video processing on free cloud resources

5. System Architecture

The system follows a modular multimodal pipeline. A user uploads a video through Streamlit. The video is saved temporarily and processed through separate pipelines for audio, vision, text, and scoring. The outputs are combined into an interactive dashboard and downloadable report.



6. Methodology

6.1 Input Video Processing

The application first receives a video file through the Streamlit file uploader. The file is stored temporarily so that OpenCV and FFmpeg-based tools can access it. The uploaded file is then displayed in the dashboard for user confirmation before analysis begins.

6.2 Audio Extraction

The system extracts the audio stream from the interview video and converts it into a mono 16 kHz WAV file. A standardized audio file is important because speech-recognition and audio-analysis libraries perform more consistently when the sample rate and channels are controlled. The application uses imageio-ffmpeg or system FFmpeg depending on the environment.

6.3 Speech Transcription

The transcription component uses Faster-Whisper in the full academic design. The model converts spoken interview answers into text segments with start and end timestamps. The transcript is important because later modules depend on text content for sentiment analysis, filler-word counting, vocabulary diversity, and expected

keyword matching. The cloud version should use the tiny model for faster execution and lower memory usage.

6.4 Voice Delivery Analysis

Librosa is used to process the extracted WAV audio. The system estimates speaking rate from the number of transcript words and audio duration. It also calculates an RMS-based volume proxy, a pitch proxy, and a silence ratio. These values are combined into a voice-delivery score. This score does not measure interview quality directly; rather, it helps the candidate observe whether the delivery was too quiet, too slow, too fast, or interrupted by excessive silence.

6.5 Text and Sentiment Analysis

The transcript is processed with regular expressions and a sentiment model. The system calculates word count, filler-word frequency, lexical diversity, sentiment polarity, and expected keyword coverage. Expected keywords are entered by the user according to the role or interview question.

6.6 Visual Attention Analysis

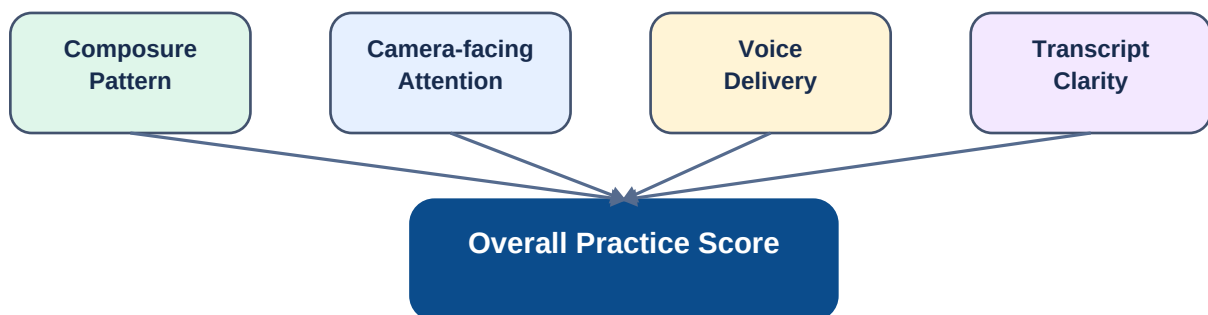
The visual-attention module samples frames from the uploaded video and uses computer vision to estimate whether the face is visible and roughly camera-facing. In the local version, the system may use MediaPipe-style face landmark logic. In the lightweight cloud version, OpenCV face detection provides a fallback. This is not a clinical eye-tracking method; it only provides an approximate camera-facing attention proxy.

6.7 Emotion Analysis

The full academic version includes a DeepFace-based emotion timeline. It samples video frames and estimates dominant facial expression categories, such as neutral, happy, sad, angry, surprise, fear, or disgust. These outputs are approximate and sensitive to lighting, camera quality, culture, facial angle, and model bias. The public Streamlit Cloud version can disable DeepFace to maintain deployment stability, while the local version can demonstrate full emotion detection.

6.8 Scoring Engine

The scoring engine combines composure pattern, camera-facing attention, voice delivery, and transcript clarity into a single overall practice score. The dashboard allows the user to adjust the weights, making the system flexible for different interview types.



7. Technology Stack

Component	Technology	Purpose
Frontend / Dashboard	Streamlit	Interactive web interface, file upload, charts, tabs, and downloads
Video Processing	OpenCV	Reading frames, sampling video, face detection fallback
Emotion Analysis	DeepFace	Local full-version facial expression timeline
Speech Recognition	Faster-Whisper	Speech-to-text transcription from interview audio
Audio Processing	Librosa and SoundFile	Voice features such as duration, RMS, pitch proxy, and silence ratio

Component	Technology	Purpose
Text Processing	Regex and TextBlob	Sentiment, filler words, keyword coverage, lexical diversity
Reporting	ReportLab	Downloadable academic PDF reports
Deployment	Streamlit Community Cloud	Public cloud demonstration of lightweight version
Version Control	GitHub	Repository hosting and cloud deployment source

8. Functional Modules

Module	Input	Processing	Output
Upload Module	Video file	Temporary save and validation	Usable video path
Audio Module	Video path	Extract audio to WAV	16 kHz mono audio file
Transcription Module	WAV file	Faster-Whisper ASR	Transcript and timestamped segments
Voice Module	WAV and word count	Duration, RMS, pitch proxy, silence ratio	Voice-delivery score
Text Module	Transcript and keywords	Sentiment, fillers, lexical diversity, keywords	Text clarity indicators
Vision Module	Video frames	Face detection and camera-facing proxy	Attention score and face visibility
Emotion Module	Sampled frames	DeepFace expression prediction	Emotion timeline and composure score
Scoring Module	All sub-scores	Weighted scoring formula	Overall practice score and performance band
Report Module	All results	PDF and CSV generation	Downloadable output files

9. Scoring Logic

The system produces individual sub-scores first, then computes a weighted overall score. The default weights are intentionally adjustable because interview contexts differ. A technical interview may focus more on content clarity and keyword coverage, while a presentation interview may focus more on delivery and camera-facing presence.

Score	Meaning	Main Signals
Composure Pattern	Stability of sampled facial-expression output	Neutral/happy ratio and emotion confidence in local DeepFace version
Camera-facing Attention	Approximate face visibility and forward-facing posture	OpenCV or landmark-based face position estimation
Voice Delivery	Speech pacing, volume proxy, pauses, and pitch proxy	Speaking rate, RMS, silence ratio, median pitch
Transcript Clarity	Language quality and relevance	Sentiment, filler words, lexical diversity, keyword coverage
Overall Practice Score	Combined practice indicator	Weighted average of all available sub-scores

10. Implementation Details

10.1 Streamlit Interface

The Streamlit interface contains a sidebar for candidate details, interview role, interview question, expected keywords, transcription model, and scoring weights. The main page contains the upload component, video preview, analysis button, metrics, charts, tabs, and download buttons. This structure makes the tool easy to use for students and evaluators.

10.2 Local and Cloud Versions

The project has two practical execution modes. The local full version can include heavier dependencies such as TensorFlow and DeepFace. The cloud-friendly version is optimized for deployment reliability and can exclude DeepFace, TensorFlow, and desktop OpenCV. This distinction is important because free cloud platforms have limited memory and system-library constraints.

10.3 Repository Structure

```
AI_Interview_Analyzer_Pro_Academic/  
  app.py  
  requirements.txt  
  README.md  
  PROJECT_REPORT_OUTLINE.md  
  check_installation.py  
  modules/ (optional modular version)  
  outputs/ (generated results)
```

11. Deployment Procedure

The project was prepared for deployment through GitHub and Streamlit Community Cloud. The recommended deployment sequence is: create a GitHub repository, upload app.py and requirements.txt, connect the repository to Streamlit Community Cloud, select Python 3.11, and deploy.

```
Local setup commands:  
cd /d "D:\DATA RECOVERY\AI_Interview_Analyzer_Pro_Academic"  
python -m venv .venv  
.venv\Scripts\activate  
python -m pip install --upgrade pip setuptools wheel  
pip install -r requirements.txt  
python -m streamlit run app.py  
  
Cloud deployment:  
Repository: twofriend/AI-Interview-Analyzer-Pro  
Branch: main  
Main file: app.py  
Python version: 3.11
```

12. Testing Strategy

Test Area	Test Case	Expected Output
File Upload	Upload a 20-40 second MP4 interview	Video preview appears and analysis button works
Audio Extraction	Video has a clear audio track	Temporary WAV file is generated
Transcription	Use tiny Whisper model	Transcript text and segments appear
Voice Analysis	Audio has speech and pauses	Duration, speaking rate, silence ratio, and delivery score appear
Text Analysis	Transcript contains expected keywords	Keyword coverage and filler-word count appear
Visual Attention	Face visible in camera	Face visibility and attention score appear
PDF Export	Click Download Detailed PDF Report	PDF report downloads successfully
Cloud Reliability	Upload short compressed video	App completes without memory crash

13. Results and Observations

The application successfully demonstrates a complete multimodal pipeline for interview-practice analysis. The most reliable cloud outputs are transcription, voice metrics, text clarity, camera-facing attention fallback, scoring, and downloadable reports. The full local version can additionally produce a DeepFace emotion timeline.

- The dashboard is accessible through a browser and can be demonstrated as a deployed application.
- The candidate can customize expected keywords according to the interview role.
- The score is not fixed; it can be adjusted through rubric weights in the sidebar.
- The application exports both machine-readable CSV and human-readable PDF reports.

- The cloud version may disable DeepFace to maintain reliability, while the local version can run the full emotion module.

14. Ethical and Responsible-Use Considerations

AI systems that analyze faces, voice, and language can be misused if their outputs are interpreted as objective truth. This project clearly labels all results as practice indicators rather than final judgments. Emotion labels are especially sensitive because facial expression models can be inaccurate across cultures, lighting conditions, disabilities, facial differences, camera angles, and recording quality.

- The system should not be used to automatically accept or reject a candidate.
- The system should not be used for psychological diagnosis, lie detection, personality evaluation, intelligence measurement, or medical interpretation.
- Users should obtain consent before uploading any real interview recording.
- Confidential interviews and resumes should not be uploaded to public cloud deployments.
- Human review must remain central in any real evaluation context.

15. Limitations

- Free Streamlit Community Cloud may not support heavy combinations of TensorFlow, DeepFace, Whisper, OpenCV, and video processing at the same time.
- DeepFace emotion labels are approximate and may not represent true emotional state.
- OpenCV face detection is sensitive to lighting, face angle, camera quality, and background.
- Speech transcription may make mistakes with accents, noise, overlapping speech, or low-quality microphones.
- Voice metrics may reflect recording quality rather than real speaking ability.
- The scoring formula is a practice rubric, not a scientifically validated employment assessment.

16. Future Enhancements

- Add role-specific question banks for data science, software engineering, cybersecurity, teaching, and HR interviews.
- Add an LLM-based answer-quality evaluator for STAR structure, technical accuracy, and relevance.
- Add resume-to-interview alignment by comparing the candidate resume with the spoken answer.
- Add multilingual transcription and feedback support.
- Add Docker deployment for the full DeepFace and TensorFlow version.
- Add secure authentication and private storage for institutional deployments.
- Add evaluator comments and side-by-side comparison across multiple interview attempts.

17. Academic Contribution

This project is academically valuable because it combines concepts from deep learning, computer vision, audio processing, speech recognition, natural language processing, software engineering, and cloud deployment. It demonstrates how a real-world AI system is not only about model selection but also about input preprocessing, dependency management, deployment constraints, user interface design, reporting, ethics, and responsible system boundaries.

18. Conclusion

AI Interview Analyzer Pro is a practical academic project that demonstrates a multimodal AI pipeline for interview preparation. It allows users to upload an interview video and receive feedback on multiple dimensions of

performance. The project shows the integration of deep learning, machine learning, audio processing, NLP, and Streamlit deployment in one application. The stable cloud version is suitable for public demonstration, while the full local version can showcase DeepFace emotion analysis. The project is best understood as an educational and portfolio tool that supports self-improvement, not as an automated recruitment decision system.

Appendix A: Recommended Cloud Requirements

```
streamlit==1.58.0
numpy==1.26.4
pandas==2.2.3
opencv-python-headless==4.10.0.84
librosa==0.10.2.post1
soundfile==0.12.1
faster-whisper==1.1.1
imageio-ffmpeg==0.6.0
textblob==0.18.0.post0
reportlab==4.2.5
Pillow==10.4.0
scipy==1.13.1
```

Appendix B: Full Local Emotion Version Dependencies

For the local full emotion-recognition version, DeepFace, TensorFlow, and tf-keras can be installed in a Python 3.11 environment. This version is heavier and should be demonstrated on a local computer or a Docker-based cloud platform with sufficient memory.

```
deepface==0.0.100
tensorflow==2.16.1
tf-keras==2.16.0
opencv-python==4.10.0.84
```